

C O N S T E L L A T I O N

Concept Paper

The Constellation Lens

A Text Network Analysis Engine for Knowledge Discovery

Inspired by InfraNodus • Engineered for Personal Knowledge Management

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1. Executive Summary

This concept paper proposes the **Constellation Lens** — a text network analysis engine for Constellation that applies network science algorithms to a user’s knowledge base, revealing topical clusters, conceptual bridge nodes, structural gaps, and cognitive diversity metrics. The concept draws direct inspiration from InfraNodus, a peer-reviewed text network analysis tool that uses graph theory to generate insight from any discourse.

Where InfraNodus analyzes individual texts or external data sources (Google results, tweets, PDFs) through a cloud-based web application, the Constellation Lens reimagines the same analytical principles for a local-first PKM context: operating across an entire universe of interconnected notes, computed in Rust on the user’s machine, and integrated with Constellation’s existing graph infrastructure.

Core proposition

The Constellation Lens transforms Constellation’s note graph from a passive relationship visualization into an active analytical engine that identifies topical communities, ranks notes by bridging importance, detects structural blind spots, and measures the cognitive diversity of a user’s knowledge — answering the question: “What patterns and gaps exist in my thinking?”

This paper is a companion to the *Constellation Map Concept Paper* (April 2026), which proposed a radial knowledge visualization inspired by Goalscape. The two concepts are complementary: the Constellation Map provides structural overview at the organizational level; the Constellation Lens provides analytical intelligence at the conceptual level. Together, they position Constellation as a knowledge intelligence platform unlike anything in the current PKM landscape.

2. Source Analysis: InfraNodus

2.1 What InfraNodus is

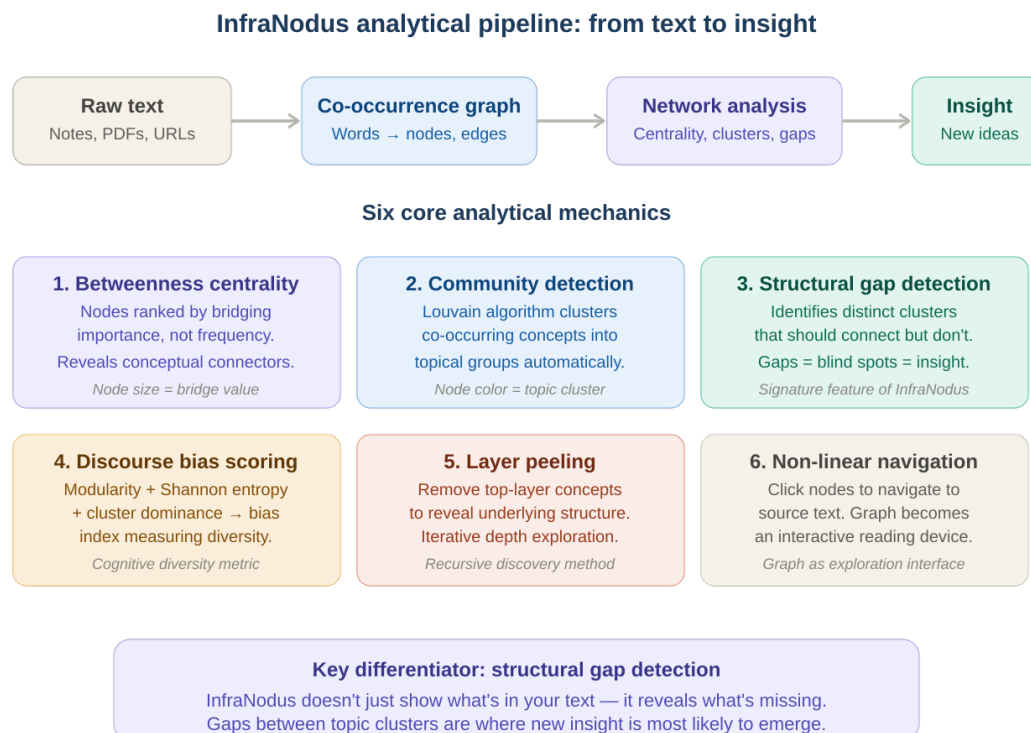
InfraNodus is a text network analysis tool developed by Dmitry Paranyushkin and Nodus Labs (based in France, registered as InfraNodus SAS). It was first prototyped in 2011, launched commercially in 2018, and published as a peer-reviewed paper at WWW’19 (The Web Conference). The tool is used by researchers, marketing professionals, writers, and organizations including Greenpeace and Procter & Gamble.

InfraNodus represents any text as a network graph where words/concepts are nodes and their co-occurrences within a sliding window create weighted edges. It then applies network science algorithms — betweenness centrality, community detection (Louvain), Force-Atlas layout — to extract structural insights that go far beyond what standard NLP, word clouds, or LDA topic modeling can provide.

Its technical stack includes NestJS (Node.js), Prisma, PostgreSQL, and Sigma.js for graph visualization, with GPT-5 integration for AI-powered insight generation. It offers an Obsidian plugin, browser extension, MCP server, API, and n8n integration.

2.2 Core analytical mechanics

The following diagram illustrates InfraNodus's analytical pipeline — the transformation from raw text to structured insight through six core mechanics:



Based on: Paranyushkin, D. (2019). InfraNodus: Generating Insight Using Text Network Analysis. WWW'19.

Figure 1: InfraNodus analytical pipeline — from raw text to insight through six network science mechanics

The six mechanics are detailed in the table below:

Mechanic	Algorithm / method	What it reveals
Betweenness centrality	For each node, count how many shortest paths between all other node pairs pass through it. Higher count = higher centrality.	The concepts that serve as bridges between different topic areas. These are not necessarily the most frequent words — they are the most structurally important connectors.
Community detection	Louvain algorithm: iteratively optimizes modularity by grouping densely-connected nodes into communities.	Topical clusters that emerge from actual co-occurrence patterns, not from user-defined tags or folders. Each cluster is color-coded on the graph.
Structural gap detection	Identify pairs of communities with high internal density but low inter-community connectivity. Highlight the shortest missing bridges.	Blind spots in a discourse: areas where two relevant topic clusters exist but lack connection. These gaps are where new insight is most likely to emerge.
Discourse bias scoring	Combine modularity ($M > 0.4$ = pronounced structure), cluster dominance (% of nodes in largest cluster), and Shannon entropy (E) of top-4 node community assignments.	A quantitative measure of whether a discourse is cognitively diverse or biased toward a narrow set of ideas. High entropy = diverse; low entropy = echo chamber.
Layer peeling	Remove the top-N nodes by centrality; recalculate all metrics on the residual graph.	The underlying structure beneath the obvious surface concepts. Iterative peeling reveals progressive depth layers.
Non-linear navigation	Click any node to retrieve all source text statements containing that concept and its neighbors.	The graph becomes an interactive reading and exploration device, not just a visualization.

2.3 The structural gap insight

InfraNodus's most distinctive contribution is the structural gap concept. Most text analysis tools focus on what's present in a text — frequent words, dominant topics, sentiment. InfraNodus inverts this by focusing on what's absent: the connections that don't exist yet between topic clusters that could meaningfully relate to each other.

The intellectual foundation draws on Ronald Burt's theory of structural holes in social networks (1992): the idea that innovation and competitive advantage arise at the boundaries between groups, not within them. InfraNodus applies this principle to knowledge: if a user's notes have a dense cluster on "Islamic jurisprudence" and a separate dense cluster on "systems engineering," but no notes connecting them, that's a structural gap — and a prompt for the user to explore whether a connection exists (e.g., how systematic legal reasoning parallels requirements engineering).

3. The Constellation Lens Concept

3.1 Definition

The **Constellation Lens** is a proposed analytical engine for Constellation that applies network science algorithms to the user's universe graph, transforming the existing GraphMind visualization from a passive map into an active knowledge discovery tool. It computes betweenness centrality, detects topical communities, identifies structural gaps, and measures cognitive diversity — all locally, in real time, within the Rust backend.

3.2 Semantic mapping: from text analysis to universe analysis

InfraNodus operates on a word-level co-occurrence graph built from a single text or corpus. The Constellation Lens operates on a note-level relationship graph built from an entire universe. The following table defines the mapping:

InfraNodus dimension	InfraNodus context	Constellation Lens context
Nodes	Words/concepts (lemmatized)	Notes (each note = one node)
Edges	Co-occurrence within a 4-gram sliding window	Wikilinks, shared tags, content similarity (TF-IDF or embedding)
Edge weight	Proximity in sentence (closer = heavier)	Link type weight: explicit wikilink > shared tag > content similarity
Betweenness centrality	Words that bridge topic clusters	Notes that bridge knowledge areas — the most structurally important notes in the universe
Community detection	Topical clusters from word co-occurrence	Knowledge domains from link/tag patterns — emergent, not folder-defined
Structural gaps	Discourse blind spots between word clusters	Knowledge blind spots: areas of the universe that should connect but don't
Bias scoring	Cognitive diversity of a single text	Cognitive diversity of the entire universe — “universe health” metric
Layer peeling	Remove top concepts to reveal depth	Hide dominant notes (e.g., MOCs, index notes) to reveal underlying conceptual structure
Non-linear navigation	Click node → source text excerpt	Click node → open note in editor (already native to GraphMind)

3.3 Three edge types for a richer graph

InfraNodus builds its graph from a single edge type (word co-occurrence). The Constellation Lens has access to a richer signal set. Three edge types are proposed:

Explicit links (weight 1.0): Wikilinks manually created by the user. These represent deliberate connections and carry the highest weight. The existing Universe Index already tracks these.

Shared tags (weight 0.6): Two notes sharing the same frontmatter tag or inline tag are implicitly related. Tag co-occurrence creates edges that the user may not have considered as connections.

Content similarity (weight 0.3): Notes with high textual similarity (computed via TF-IDF cosine similarity or, optionally, embedding similarity) but no explicit link between them. These are the most interesting edges because they reveal connections the user hasn't yet made explicit. Content similarity is computed lazily and cached.

The multi-edge approach means the Constellation Lens can detect structural gaps even when the user hasn't created wikilinks — surfacing “you wrote about similar topics in these two notes but never linked them” as an actionable suggestion.

3.4 Universe health: the cognitive diversity metric

Adapting InfraNodus's discourse bias scoring, the Constellation Lens computes a composite “universe health” metric from three inputs:

Component	Metric	What it measures	Healthy range
Modularity (M)	Louvain modularity score	How distinct the topical clusters are. $M > 0.4$ indicates well-defined communities.	0.3 – 0.6
Dominance (D)	% of notes in the largest community	Whether one topic overwhelms the universe. $D > 50\%$ suggests imbalance.	< 35%
Entropy (E)	Shannon entropy of community distribution	How evenly knowledge is distributed across topics. Higher = more diverse.	> 2.0 bits
Connectivity (C)	Average path length between communities	How well-bridged the knowledge areas are. Low = well-connected.	< 4 hops

These components combine into a single diversity score (0–100) displayed as a compact indicator in the GraphMind sidebar. A universe with high modularity, low dominance, high entropy, and low inter-community path length is a cognitively diverse, well-bridged knowledge base.

4. Design Principles

Principle 1: Reveal, don't prescribe

The Lens surfaces patterns and gaps; it does not tell the user what to do. A structural gap is information, not an instruction. The user decides whether to bridge it, ignore it, or explore it later.

Principle 2: Compute locally, present unobtrusively

All graph analytics run in the Rust backend on the user's machine. No data leaves the device. Results appear as overlays and sidebar panels within GraphMind, not as a separate tool or modal. The user can toggle analytics on/off without leaving their workflow.

Principle 3: Multiple edge types, single coherent graph

Wikilinks, shared tags, and content similarity are combined into one weighted graph. The user doesn't need to understand graph theory — they see one visualization where thickness and opacity of edges reflect connection strength, and colors reflect automatically-detected topic clusters.

Principle 4: Emergent structure over imposed structure

Community detection produces clusters from actual content patterns, not from the user's folder hierarchy. This respects the bottom-up ethos of Zettelkasten and Digital Gardening while providing the top-down overview that users crave.

Principle 5: Language-agnostic by design

The graph analytics layer operates on structural relationships (links, tags, similarity scores), not on language-specific NLP. This means the Constellation Lens works identically for Arabic, English, or bilingual universes — a critical requirement for Constellation's RTL-first architecture. Language-specific processing (lemmatization, stopword removal) is only needed for the optional content similarity edge, where it can be modularized per language.

5. Use Cases

5.1 Knowledge blind spot discovery

A researcher studying both Islamic intellectual heritage and systems engineering has dense clusters of notes in each domain. The Lens detects a structural gap between them and surfaces it as: “These two knowledge areas are well-developed but disconnected. Consider whether concepts like *maqasid al-shari’a* (objectives-based reasoning) relate to requirements traceability in SE.” The researcher creates a bridging note — and a new line of inquiry emerges.

5.2 Bridge note identification

A user has a note titled “Epistemology” that links to both philosophy and AI/ML notes. Betweenness centrality ranks it as the #3 most structurally important note in the universe, even though it has fewer links than heavily-connected index notes. The user realizes this note is a critical conceptual junction and invests in developing it further.

5.3 Universe health audit

A student checks their universe health metric and sees: Modularity 0.7, Dominance 62%, Entropy 1.4. The Lens flags this as a low-diversity universe dominated by one topic. The student uses this as motivation to explore underdeveloped areas, using the structural gaps as starting points.

5.4 Research literature mapping

A PhD candidate imports 50 research paper notes into their universe. The Lens automatically clusters them into topical communities and reveals which sub-topics are well-covered and which represent gaps in their literature review. Content similarity edges highlight papers that discuss related concepts but that the candidate hasn’t yet cross-referenced.

5.5 Layer peeling for deep exploration

A user’s universe graph is dominated by a few heavily-linked MOC notes. They activate layer peeling, which hides the top-10 centrality nodes and recalculates the graph. Suddenly, a secondary structure emerges: smaller conceptual clusters and bridge notes that were invisible under the MOC layer. This is analogous to removing the obvious to reveal the subtle — a principle that resonates with the *tawil* (تأويل) tradition of reading beneath surface meaning.

6. Integration Architecture

6.1 Position within Constellation

The Constellation Lens is not a separate view — it is an analytical layer within the existing GraphMind graph view. When the user activates the Lens, the graph gains overlay features: community coloring, centrality-based node sizing, gap highlighting, and a sidebar analytics panel. The Lens enhances GraphMind; it does not replace it.

Component	Engine	Without Lens	With Lens active
Node sizing	Pixi.js (WebGL)	Sized by link count (degree)	Sized by betweenness centrality (bridging importance)
Node coloring	Pixi.js	User-defined or uniform	Auto-colored by detected community
Edge rendering	Pixi.js	Wikilinks only	Wikilinks + shared tags + content similarity (toggleable)
Sidebar	Svelte 5	Basic note metadata	Analytics panel: top communities, top bridges, structural gaps, universe health
Interaction	Pixi.js events	Click to open note	Click to open note + select to explore neighborhood + right-click for gap suggestions

6.2 Computation pipeline

1. Universe Index (Rust): maintains the note graph with wikilinks and metadata. Extended to compute shared-tag edges and, optionally, content similarity edges (TF-IDF vectors per note, cosine similarity with configurable threshold).
2. Graph analytics engine (Rust): a new module implementing betweenness centrality (Brandes' algorithm, $O(VE)$), Louvain community detection, modularity scoring, Shannon entropy, and structural gap identification. Runs asynchronously; results cached and invalidated on universe change.
3. IPC bridge (Tauri v2): exposes `constellation_lens_data` command returning a JSON payload with per-node centrality, community assignment, and a list of detected structural gaps as node-pair suggestions.
4. GraphMind overlay (Svelte 5 + Pixi.js): consumes the JSON and applies visual overlays — community colors, centrality sizing, gap highlighting (dashed lines between unconnected community pairs), and the sidebar analytics panel.

6.3 Performance considerations

Brandes' algorithm for betweenness centrality has time complexity $O(VE)$, which for a universe of 5,000 nodes with 20,000 edges means approximately 100 million operations — trivial for a Rust implementation running on modern hardware (sub-second). Louvain community detection is $O(n \cdot \log(n))$ and similarly fast. Content similarity computation is the most expensive operation and should be incremental: compute vectors only for changed/new nodes, cache all vectors, and recompute similarities lazily.

For universes exceeding 50,000 nodes, approximate betweenness centrality (sampling-based) can be used without significant accuracy loss, as demonstrated in the network science literature.

7. Synergy with the Constellation Map

The Constellation Lens and the Constellation Map (proposed in the companion concept paper) are complementary analytical tools that answer different questions about the same knowledge base:

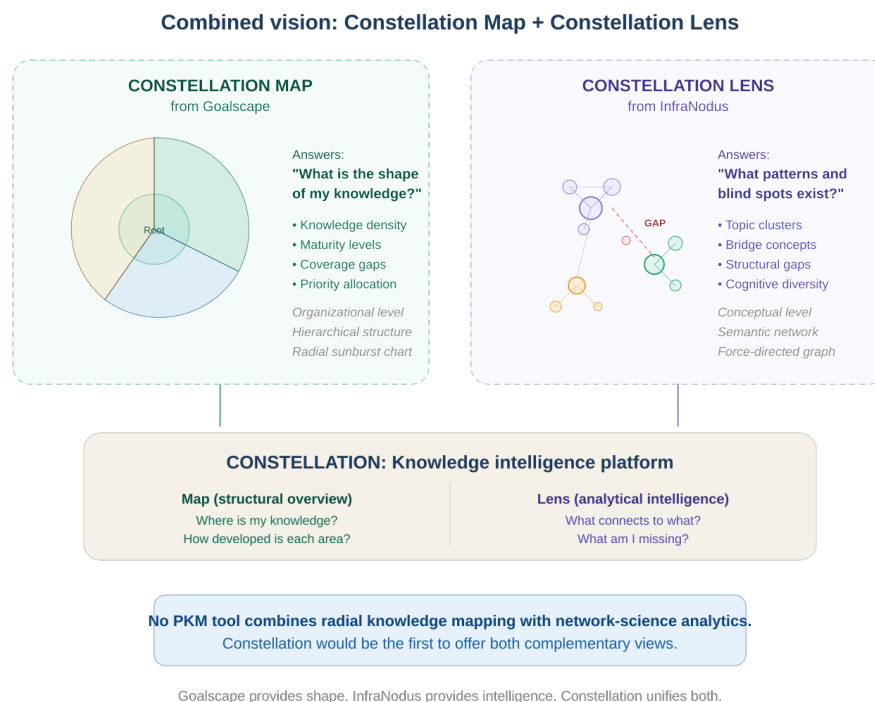


Figure 2: Combined vision — the Constellation Map provides shape; the Constellation Lens provides intelligence

Dimension	Constellation Map	Constellation Lens
Source inspiration	Goalscape (goal management)	InfraNodus (text network analysis)
Chart type	Radial sunburst (concentric rings)	Force-directed network graph
Primary question	What is the shape and maturity of my knowledge?	What patterns and blind spots exist in my thinking?
Analytical level	Organizational (areas, topics, folders)	Conceptual (semantic relationships, co-occurrence)
Node meaning	Knowledge branch (subtree of notes)	Individual note
Key metric: size	Knowledge weight (density, volume)	Betweenness centrality (bridging importance)
Key metric: color	Categorical grouping (user-defined)	Community membership (auto-detected)
Key metric: fill	Maturity (seed → evergreen)	N/A (not hierarchical)
Unique capability	Zero-sum priority allocation	Structural gap detection
Computation backend	Rust (Universe Index subtree stats)	Rust (graph analytics engine)
Rendering	SVG via D3.js	Canvas via Pixi.js (GraphMind)

The data flow between them is complementary: both read from the same Universe Index, but compute different analytics. A user might open the Constellation Map to see that their “Systems Engineering” branch is underdeveloped (thin segment, low maturity), then switch to the Constellation Lens to discover which specific notes in that area would benefit from connections to other domains (structural gaps). The Map diagnoses; the Lens prescribes.

8. Competitive Differentiation

Application	Graph type	Centrality ranking	Community detection	Structural gap detection	Universe health metric
Obsidian	Force-directed	No (degree only)	No	No	No
Logseq	Force-directed	No	No	No	No
Roam Research	Force-directed	No	No	No	No
Notion	None	N/A	No	No	No
InfraNodus	Force-directed	Yes (betweenness)	Yes (Louvain)	Yes	Yes (discourse bias)
InfraNodus + Obsidian plugin	Force-directed	Yes	Yes	Yes	Partial
Constellation (proposed)	Force-directed + radial	Yes (betweenness)	Yes (Louvain)	Yes	Yes (universe health)

InfraNodus's Obsidian plugin provides some of these capabilities but operates as an **external SaaS service**: notes are uploaded to InfraNodus's EU servers for processing. Constellation's advantage is **local-first computation**: all analytics run on the user's machine in Rust, with no data leaving the device. This is a fundamental architectural difference for privacy-conscious users.

Additionally, InfraNodus analyzes text at the word level (co-occurrence of terms). The Constellation Lens analyzes at the note level (relationships between documents), which is the natural unit for PKM. This means the Lens can leverage structural signals (wikilinks, tags, library, folder hierarchy) that InfraNodus's word-level approach cannot access.

9. Risks and Mitigations

Risk	Severity	Mitigation
Information overload from too many suggested gaps	Medium	Limit displayed gaps to top-5 by default. Rank gaps by a relevance score combining community importance and gap width. Let users dismiss suggestions.
Betweenness centrality biased toward MOC/index notes	Medium	Layer peeling toggle: let users exclude high-degree nodes from centrality computation. Alternatively, normalize centrality by degree.
Content similarity is computationally expensive	Medium	Make it opt-in. Compute incrementally on changed notes only. Cache TF-IDF vectors. Use approximate methods for large universes.
Users may not understand graph analytics terminology	Low	Use plain-language labels: “Bridge notes” not “Betweenness centrality.” “Knowledge clusters” not “Communities.” “Blind spots” not “Structural gaps.”
Arabic text requires specific lemmatization for content similarity	Low	Use a modular NLP pipeline with language-specific stemmers. Arabic stemming (ISRISemmer or similar) as a configurable option. Tag/link-based edges work without any NLP.
Feature overlap with GraphMind	Low	The Lens is an overlay on GraphMind, not a replacement. GraphMind without the Lens remains a valid, simpler view. The Lens adds optional depth.

10. Suggested Development Roadmap

Phase 1: Graph analytics engine (foundation)

- Implement Brandes’ betweenness centrality algorithm in Rust.
- Implement Louvain community detection in Rust.
- Expose analytics via IPC (constellation_lens_data command).
- Apply community coloring and centrality sizing as overlays in GraphMind.
- Deliverable: GraphMind nodes auto-colored by topic, sized by importance.

Phase 2: Structural gaps and sidebar (intelligence)

- Implement structural gap detection (inter-community distance analysis).
- Build the analytics sidebar panel in Svelte 5: top communities, top bridges, top gaps.
- Add gap highlighting in the graph (dashed lines between unconnected community pairs).
- Compute universe health metric (modularity + dominance + entropy + connectivity).
- Deliverable: an active knowledge discovery tool with actionable suggestions.

Phase 3: Multi-edge and layer peeling (depth)

- Add shared-tag edges as a second edge type (toggleable).
 - Implement content similarity edges (TF-IDF, incremental computation, opt-in).
 - Add layer peeling: hide top-N centrality nodes and recalculate.
 - Integrate with the Constellation Map: allow clicking a Map segment to open the Lens filtered to that knowledge branch.
 - Deliverable: a comprehensive, dual-view knowledge intelligence platform.
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11. Conclusion

InfraNodus demonstrates that applying network science to text — specifically betweenness centrality, community detection, and structural gap analysis — produces insights that no amount of keyword counting, sentiment analysis, or LLM summarization can replicate. Its peer-reviewed methodology has been validated by researchers, organizations, and thousands of users worldwide.

The Constellation Lens proposes to transplant this analytical paradigm from cloud-based text analysis to local-first personal knowledge management, operating at the note level rather than the word level, and leveraging Constellation’s existing Rust backend and GraphMind infrastructure. The result is a knowledge discovery engine that transforms a passive graph visualization into an active partner in thinking.

Combined with the Constellation Map (proposed in the companion paper), the Lens completes Constellation’s vision as a knowledge intelligence platform: the Map shows the shape of knowledge; the Lens reveals its hidden patterns. No competing PKM tool offers either capability, let alone both.

Next steps

1. Review and approve this concept paper alongside the Constellation Map paper.
2. Design the Rust graph analytics module API (betweenness centrality, Louvain, gap detection).
3. Prototype community coloring and centrality sizing as a GraphMind overlay.
4. Validate performance on a test universe of 5,000+ notes.
5. Define the IPC contract and Svelte 5 sidebar component specification.

End of concept paper — Constellation Lens v1.0